Quiz Review

Name: _____ Date: _____

Part 1: Distributive Property

Example: Use the distributive property to simplify 4(x + 3).

Multiply the 4 by each term inside the parentheses:

4(x + 3)

4 · x = 4x

4 · 3 = 12

4x + 12

Your turn: Use the distributive property to simplify each expression.

1. 5(x + 2)

2. 3(x − 4)

3. 2(3x + 5)

4. 4(2x − 3)

Part 2: Variables on Both Sides

Example: Solve $5x + 2 = 3x + 10$.

Step 1: Subtract $3x$ from both sides to get the variables on one side: $2x + 2 = 10$

Step 2: Subtract 2 from both sides: $2x = 8$

Step 3: Divide both sides by 2: **$x = 4$**

Your turn: Solve each equation. Show your steps.

1. $4x + 3 = 2x + 9$

2. $6x - 5 = 2x + 7$

3. $7x + 1 = 3x + 17$

4. $5x - 4 = 3x + 6$

Part 3: Multistep Equations

Example: Solve 2(x + 3) = 14.

Step 1: Distribute the 2: 2x + 6 = 14

Step 2: Subtract 6 from both sides: 2x = 8

Step 3: Divide both sides by 2: **x = 4**

Your turn: Solve each equation. Show your steps.

1. 4x − 7 = 21 + 6x

2. 3(x + 2) = 24

Answer Key (Teacher Copy)

Part 1: 1) $5x + 10$ 2) $3x - 12$ 3) $6x + 10$ 4) $8x - 12$

Part 2: 1) $x = 3$ 2) $x = 3$ 3) $x = 4$ 4) $x = 5$

Part 3: 1) $x = -14$ 2) $x = 6$